

Robotic Manipulators

Force Control

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Version: April 2026

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Chapter 8

Force Control of Robotic Manipulators

This chapter aims to introduce the fundamental concepts and methods for controlling the interaction between robotic manipulators and their environment. In many manipulation tasks, it is not sufficient to control motion alone; instead, the robot must regulate the forces arising from contact with the environment.

The chapter begins by examining the nature of manipulator–environment interaction and highlights the limitations of purely motion-based control. To address these challenges, inverse dynamics control is reformulated in operational space, providing a foundation for designing controllers that directly regulate task-space behavior.

Building on this framework, the concept of mechanical impedance is introduced, leading to impedance control strategies that shape the dynamic relationship between motion and force. Both implementations with and without force measurements are examined, illustrating how compliant behavior can be achieved in practice. Furthermore, commonly used force and torque sensing technologies are presented.

The chapter then presents direct force control methods, where the interaction force is explicitly regulated through feedback. Different control architectures are explored, including force control with an inner position loop, which improves stability by leveraging existing motion control structures.

Finally, parallel force/position control is introduced to enable the simultaneous regulation of motion and force in different directions, particularly in the presence of task constraints. This provides a unified framework for handling constrained interaction tasks commonly encountered in robotic applications.

The intended learning outcomes are as follows:

- understand the challenges associated with controlling the interaction between a robot

and its environment;

- explain the difference between motion control and force control in robotic systems;
- describe common approaches to interaction control, including impedance control and direct force control;
- understand how task-space control can be used to regulate both motion and force;
- explain how constraints affect robot behavior during contact tasks;
- describe methods for achieving simultaneous control of motion and force in robotic manipulation;
- apply appropriate control strategies to different types of interaction tasks.

8.1 Introduction

One of the fundamental requirements for the successful execution of manipulation tasks is the ability to properly handle the interaction between the manipulator and its environment. In many practical applications, such as assembly, surface finishing, and human-robot interaction, the robot is required not only to follow a desired motion but also to regulate the forces arising from contact.

The key quantity that characterizes this interaction is the *contact force* at the end-effector. Excessive contact forces are generally undesirable, as they may lead to damage of both the manipulator and the environment. Therefore, effective control of interaction forces is essential for safe and reliable operation.

In this lecture, we study how motion control strategies can be extended to account for interaction with the environment. We first introduce the concepts of *mechanical compliance* and *impedance*, which describe how the manipulator responds to external forces. These concepts provide a systematic way to incorporate force information into the control design.

We then present *force control* schemes, which are typically obtained by augmenting motion control architectures with an outer force feedback loop.

8.2 Manipulator Interaction with Environment

Control of interaction between a robot manipulator and the environment is essential for the successful execution of many practical tasks, such as polishing, deburring, machining, and

assembly. In these applications, the end-effector must not only follow a desired motion but also interact safely and effectively with the environment.

During interaction, the environment imposes constraints on the motion of the end-effector. This situation is referred to as *constrained motion*. In such cases, a purely motion-based control strategy is generally insufficient, since contact with the environment generates forces that influence the system behaviour.

In principle, successful interaction using motion control alone would require an accurate model of both the manipulator (kinematics and dynamics) and the environment (geometry and mechanical properties). While robot models can be obtained with reasonable precision, an accurate description of the environment is often difficult or impractical to achieve.

As a result, small modelling or planning errors can lead to unintended contact forces. These forces may cause deviations from the desired trajectory, and the control system may react by increasing actuator effort to compensate for the error. This can result in large contact forces, potentially leading to actuator saturation or damage to the manipulator or the environment.

This problem becomes more severe when the environment is stiff and when high-precision position control is used. To overcome these limitations, it is necessary to introduce compliant behaviour in the interaction.

From this perspective, the *contact force* becomes the key quantity describing the interaction between the manipulator and the environment. The availability of force measurements enables improved control performance in contact tasks.

Based on how force is incorporated into the control design, interaction control strategies can be broadly classified into two categories:

- *Indirect force control*, where force is regulated through motion control without explicitly closing a force feedback loop (e.g., impedance control);
- *Direct force control*, where the contact force is explicitly regulated using a force feedback loop.

In the following sections, we first introduce inverse dynamics control in operational space, which serves as the foundation for interaction control. We then present impedance control as a representative indirect force control strategy, followed by direct force control methods.

8.3 Inverse Dynamics Control in Operational Space

In the previous chapter, motion control was formulated in the joint space. However, in many robotic applications, interaction forces between the manipulator and the environment are generated in the *task space* (or operational space). Therefore, it is necessary to consider robot control design directly in operational space.

In this section, we introduce inverse dynamics control formulated in operational space. This framework will serve as the foundation for the force control methods developed in this chapter, including impedance control and direct force control.

Let $x \in \mathbb{R}^m$ denote the task-space variable (e.g., end-effector position), which is related to the joint coordinates $q \in \mathbb{R}^n$ through the forward kinematics:

$$x = f(q). \quad (8.1)$$

Differentiating with respect to time yields

$$\dot{x} = J(q)\dot{q}, \quad (8.2)$$

$$\ddot{x} = J(q)\ddot{q} + \dot{J}(q, \dot{q})\dot{q}, \quad (8.3)$$

where $J(q)$ is the Jacobian matrix.

Control Objective Given a desired task-space trajectory $x_d(t)$, the goal is to design a control input u such that

$$\lim_{t \rightarrow \infty} (x_d(t) - x(t)) = 0. \quad (8.4)$$

Define the task-space tracking error as

$$\tilde{x} = x_d - x. \quad (8.5)$$

Task-Space Error Dynamics Design We aim to impose a desired second-order error dynamics of the form

$$\ddot{\tilde{x}} + K_D \dot{\tilde{x}} + K_P \tilde{x} = 0, \quad (8.6)$$

where $K_P \in \mathbb{R}^{6 \times 6}$ and $K_D \in \mathbb{R}^{6 \times 6}$ are positive definite gain matrices. To simplify the notation, the same variables K_P and K_D are used as in the previous chapter on joint-space motion control. In task-space control, the dimension is typically six, corresponding to the end-effector pose, whereas the dimension of the joint space depends on the number of joints of the robot.

To achieve this, define the desired task-space acceleration as

$$\ddot{x} = \ddot{x}_d + K_D \dot{\tilde{x}} + K_P \tilde{x}. \quad (8.7)$$

Substituting it into the kinematic relation eq. (8.3) gives

$$J(q)\ddot{q} + \dot{J}(q, \dot{q})\dot{q} = \ddot{x}_d + K_D \dot{\tilde{x}} + K_P \tilde{x}. \quad (8.8)$$

Solving for $\ddot{q}$ yields

$$\ddot{q} = J^{-1}(q) \left(\ddot{x}_d - \dot{J}(q, \dot{q})\dot{q} + K_D \dot{\tilde{x}} + K_P \tilde{x} \right). \quad (8.9)$$

Define the auxiliary input

$$a_q = J^{-1}(q) \left(\ddot{x}_d - \dot{J}(q, \dot{q})\dot{q} + K_D \dot{\tilde{x}} + K_P \tilde{x} \right). \quad (8.10)$$

Inverse Dynamics Control Law Using the joint-space inverse dynamics formulation, the control input is given by

$$u = D(q)a_q + C(q, \dot{q})\dot{q} + G(q). \quad (8.11)$$

Substituting a_q into this expression yields the operational space inverse dynamics controller:

$$u = D(q)J^{-1}(q) \left(\ddot{x}_d - \dot{J}(q, \dot{q})\dot{q} + K_D \dot{\tilde{x}} + K_P \tilde{x} \right) + C(q, \dot{q})\dot{q} + G(q). \quad (8.12)$$

Closed-Loop Behavior By construction, this control law enforces the desired task-space error dynamics:

$$\ddot{\tilde{x}} + K_D \dot{\tilde{x}} + K_P \tilde{x} = 0. \quad (8.13)$$

Thus, the task-space tracking error converges asymptotically to zero, and the end-effector follows the desired trajectory. This formulation will serve as the basis for interaction control strategies developed later in this chapter, where task-space acceleration is shaped to regulate contact forces.

Remarks

- This approach extends inverse dynamics control from joint space to task space, allowing direct control of physically meaningful quantities such as end-effector position.
- The method requires computation of the Jacobian $J(q)$ and its derivative $\dot{J}(q, \dot{q})$.
- The inverse of $J(q)$ exists only when the manipulator is not in a singular configuration and $J(q)$ is a square matrix. Near singularities or for non-square $J(q)$, alternative formulations (e.g., pseudoinverse or damped least squares) are required.

- Compared to joint-space control, task-space control provides more intuitive behavior for tasks involving interaction with the environment.

8.4 Force/Torque Sensing in Robotic Manipulation

Accurate measurement of interaction forces is a key requirement for implementing force control strategies. In practice, different sensing technologies are used depending on the application, cost, and required accuracy.

In this section, we introduce three commonly used approaches for force/torque sensing in robotic systems: 1) Wrist force/torque sensors; 2) Joint torque sensors; 3) Tactile sensors.

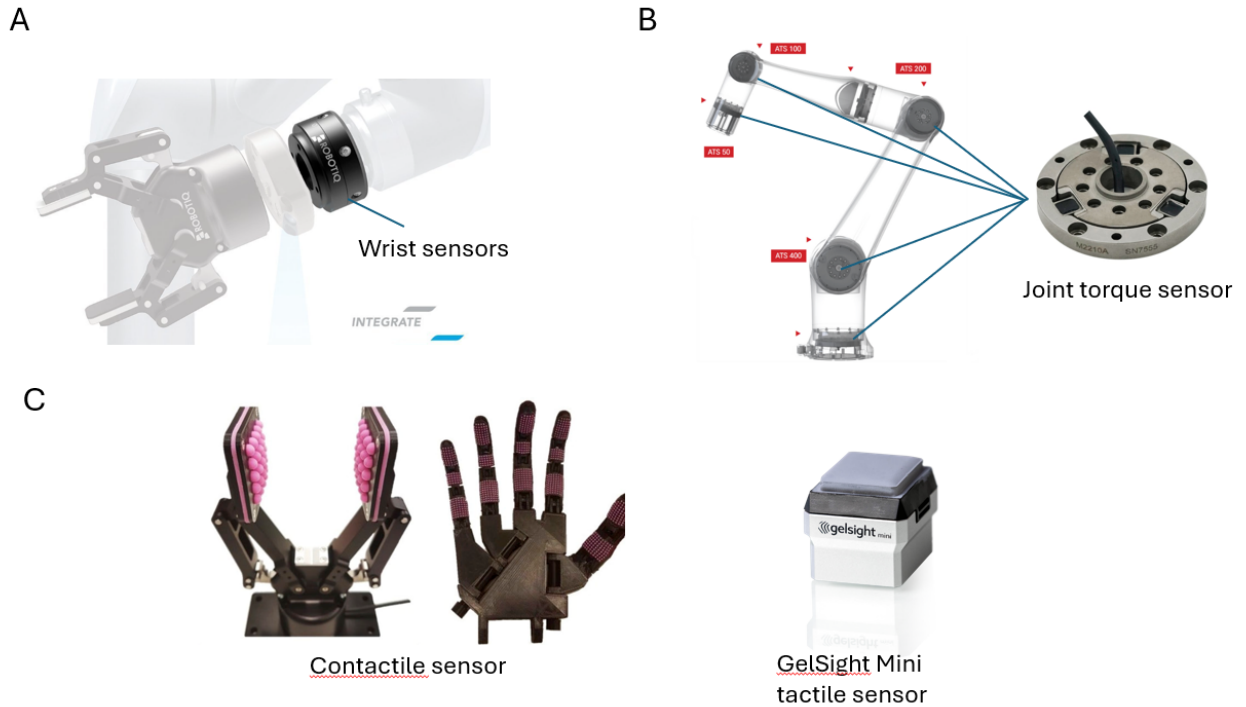


Figure 8.1: Different types of force sensors. A) Wrist force/torque sensors; B) Joint torque sensors; C) Tactile sensors.

Wrist Force/Torque Sensors

Wrist force/torque sensors are typically mounted between the robot flange and the end-effector. They directly measure the interaction forces and torques at the contact point.

- Provide full 6-axis measurements (3 forces and 3 torques);
- Located close to the end-effector, enabling accurate sensing of contact interactions;

- Widely used in force control, impedance control, and human-robot interaction;

Wrist sensors provide the most direct and accurate measurement of interaction forces. However, they increase system cost and may reduce mechanical robustness.

Joint Torque Sensors

Joint torque sensors measure the torque at each joint of the manipulator. These measurements can be used to estimate the end-effector force through the Jacobian relationship:

$$\tau = J^T(q)F_e. \quad (8.14)$$

- Provide torque measurements at each joint;
- Commonly integrated in collaborative robots (e.g., Kinova Gen3);
- End-effector forces can be estimated using robot kinematics;
- No additional hardware is required at the end-effector.

Joint torque sensing provides a compact and integrated solution. However, the estimation of end-effector forces depends on the accuracy of the robot model and may be affected by friction and unmodeled dynamics.

Tactile Sensors

Tactile sensors measure distributed contact information over a surface, such as pressure or deformation. They are typically mounted on robot grippers or fingertips.

- Provide spatially distributed contact information;
- Can detect contact location, pressure distribution, and slip;
- Useful for grasping, manipulation, and dexterous tasks;
- Examples include Contactile sensors and GelSight tactile arrays.

Unlike wrist or joint sensors, tactile sensors provide rich local contact information. They are particularly useful for manipulation tasks involving uncertainty, but are typically more complex to process and integrate.

The choice of force sensing technology depends on the task requirements:

- Wrist sensors: high accuracy and direct force measurement;
- Joint torque sensors: integrated solution with model-based estimation;
- Tactile sensors: rich contact information for manipulation.

In practice, these sensing modalities can be combined to achieve robust and reliable interaction control. In the following sections, we will consider impedance control strategies both with and without contact force measurements.

When force measurements are available, we assume that a wrist-mounted force/torque sensor is used to directly obtain the full six-dimensional interaction wrench (three forces and three torques) at the end-effector. This enables explicit incorporation of the measured contact force into the control law. In the absence of force sensing, the controller must rely on the robot dynamics and motion feedback to achieve an implicit impedance behavior, as will be discussed in the subsequent sections.

8.5 Impedance Control in Operational Space

8.5.1 Mechanical Impedance

Before introducing the control design, we first introduce the concept of *mechanical impedance*. The dynamic behaviour of a mechanical system can be described by a second-order model:

$$M_d \ddot{x} + K_D \dot{x} + K_P x = F_e, \quad (8.15)$$

where M_d , K_D , and K_P denote the desired mass, damping, and stiffness of the system, respectively, and F_e is the external force applied to the system.

This equation defines a relationship between motion and force, and is referred to as the *mechanical impedance* of the system. It characterizes how the system reacts to external forces. The key idea of impedance control is to make the robot behave like such a compliant mechanical system during interaction with the environment.

8.5.2 Impedance Control with Interaction Force Measurements

Joint Space Dynamics Model

To design an impedance controller, we start from the dynamic model of the manipulator:

$$D(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = u + J^T(q)F_e, \quad (8.16)$$

where $F_e \in \mathbb{R}^6$ denotes the external forces and torques applied at the end-effector.

Inverse Dynamics Control

We consider an inverse dynamics control law with explicit compensation of the external force:

$$u = D(q)a_q + C(q, \dot{q})\dot{q} + G(q) - J^T(q)F_e. \quad (8.17)$$

Substituting into the dynamics yields a fully linearized system:

$$\ddot{q} = a_q. \quad (8.18)$$

This shows that joint acceleration can be directly assigned through the control input a_q , similar to motion control using inverse dynamics in operational space

Task Space Kinematics

The end-effector velocity and acceleration are given by:

$$\dot{x} = J(q)\dot{q}, \quad \ddot{x} = J(q)\ddot{q} + \dot{J}(q, \dot{q})\dot{q}. \quad (8.19)$$

Substituting $\ddot{q} = a_q$:

$$\ddot{x} = J(q)a_q + \dot{J}(q, \dot{q})\dot{q}. \quad (8.20)$$

Design of the Control Input

We define the tracking error:

$$\tilde{x} = x_d - x. \quad (8.21)$$

To impose a desired impedance behaviour, we specify the following second-order dynamics:

$$M_d\ddot{\tilde{x}} + K_D\dot{\tilde{x}} + K_P\tilde{x} = F_e, \quad (8.22)$$

which describes a mass-spring-damper system driven by the external force.

Expanding $\ddot{\tilde{x}}$:

$$\ddot{\tilde{x}} = \ddot{x}_d - \ddot{x} = \ddot{x}_d - Ja_q - \dot{J}\dot{q}. \quad (8.23)$$

Substituting into the impedance equation gives:

$$M_d(\ddot{x}_d - Ja_q - \dot{J}\dot{q}) + K_D\dot{\tilde{x}} + K_P\tilde{x} = F_e. \quad (8.24)$$

Rearranging, we obtain:

$$Ja_q = \ddot{x}_d + M_d^{-1}(K_D\dot{\tilde{x}} + K_P\tilde{x} - F_e) - \dot{J}\dot{q}. \quad (8.25)$$

Assuming J is invertible, the control input is chosen as:

$$a_q = J^{-1} \left[\ddot{x}_d + M_d^{-1}(K_D\dot{\tilde{x}} + K_P\tilde{x} - F_e) - \dot{J}\dot{q} \right]. \quad (8.26)$$

Impedance Behaviour

With this choice of a_q , the closed-loop system satisfies:

$$M_d\ddot{\tilde{x}} + K_D\dot{\tilde{x}} + K_P\tilde{x} = F_e, \quad (8.27)$$

which corresponds to a generalized mechanical impedance.

Therefore, the manipulator behaves as a mass-spring-damper system interacting with the environment.

Remarks

- Impedance control allows shaping the dynamic interaction between the robot and the environment. Comparing the error dynamics of impedance control in eq. (8.27) with those of inverse dynamics control in operational space in eq. (8.13), it can be observed that impedance control allows a steady-state error that is proportional to the external force, whereas inverse dynamics control tends to drive the error to zero, potentially requiring large actuator torques.
- The parameters M_d , K_D , and K_P determine the desired behaviour. The desired interaction force F_d is achieved by adjusting these desired impedance parameters and desired end-effector pose x_d .
- The approach relies on accurate modelling and force measurements.

8.5.3 Impedance Control without Interaction Force Measurements

In many practical situations, force measurements may not be available. In this case, the controller cannot explicitly compensate or regulate the external force. Instead, the interaction behaviour must emerge from the closed-loop dynamics of the system.

Joint Space Dynamics Model

Consider the inverse dynamics control law without force compensation:

$$u = D(q)a_q + C(q, \dot{q})\dot{q} + G(q). \quad (8.28)$$

Substituting into the manipulator dynamics:

$$D(q)\ddot{q} = D(q)a_q + J^T(q)F_e, \quad (8.29)$$

which leads to:

$$\ddot{q} = a_q + D^{-1}(q)J^T(q)F_e. \quad (8.30)$$

This shows that the external force is not cancelled, but directly affects the joint acceleration.

Task Space Dynamics

Substituting $\ddot{q}$ task space acceleration in eq. (8.19):

$$\ddot{x} = J(q)a_q + \dot{J}(q, \dot{q})\dot{q} + J(q)D^{-1}(q)J^T(q)F_e. \quad (8.31)$$

Control Design

Using a structure similar to eq. (8.26), choose the control input as:

$$a_q = J^{-1} \left[\ddot{x}_d + M_d^{-1}(K_D\dot{\tilde{x}} + K_P\tilde{x}) - \dot{J}\dot{q} \right]. \quad (8.32)$$

Substituting into $\ddot{x}$ yields:

$$\ddot{x} = \ddot{x}_d + M_d^{-1}(K_D\dot{\tilde{x}} + K_P\tilde{x}) + JD^{-1}J^TF_e. \quad (8.33)$$

Closed-Loop Behaviour

Using $\ddot{\tilde{x}} = \ddot{x}_d - \ddot{x}$, we obtain:

$$\ddot{\tilde{x}} = -M_d^{-1}(K_D\dot{\tilde{x}} + K_P\tilde{x}) - JD^{-1}J^T F_e. \quad (8.34)$$

Multiplying both sides by M_d :

$$M_d\ddot{\tilde{x}} + K_D\dot{\tilde{x}} + K_P\tilde{x} = -M_dJD^{-1}J^T F_e. \quad (8.35)$$

Interpretation

Defining the operational space inertia:

$$D_A(q) = J^{-T}(q)D(q)J^{-1}(q), \quad (8.36)$$

we note that:

$$JD^{-1}(q)J^T = D_A^{-1}(q). \quad (8.37)$$

Therefore, the closed-loop system becomes:

$$M_d\ddot{\tilde{x}} + K_D\dot{\tilde{x}} + K_P\tilde{x} = -M_dD_A^{-1}(q)F_e. \quad (8.38)$$

This equation shows that a mechanical impedance behaviour naturally emerges, even without explicit force measurement. The external force influences the system through the robot dynamics.

Compared to impedance control with force compensation:

- The interaction force is not explicitly controlled;
- The resulting impedance behaviour depends on the robot dynamics through $D_A(q)$ and M_d ;
- The system exhibits an *implicit impedance*, determined by both the controller and the robot model.

8.6 Direct Force Control

8.6.1 Motivation and Concept

The interaction control schemes introduced so far, such as impedance control, belong to the class of *indirect force control*. In these approaches, the contact force is not directly regulated, but instead arises from the motion of the manipulator interacting with the environment.

However, in many practical applications, such as assembly, polishing, or machining, it is required to accurately regulate the contact force to a desired value. In these cases, indirect control may not be sufficient, and it becomes necessary to explicitly control the interaction force.

Direct Force Control Framework

The objective of direct force control is to enforce:

$$F_e \rightarrow F_d, \quad (8.39)$$

where F_d is the desired contact force and F_e is the measured force at the end-effector.

A typical force control scheme is based on a feedback loop. The force error is defined as:

$$\tilde{F} = F_d - F_e, \quad (8.40)$$

and is used to generate a control action through a force controller.

In particular, the control input is constructed as:

$$\tau = f(\tilde{F}), \quad (8.41)$$

which is then mapped into joint torques through the Jacobian transpose.

Force Feedback Control Law

A simple and commonly used implementation is a proportional force controller:

$$\tau = J^T K_F \tilde{F}, \quad (8.42)$$

where K_F is a positive definite gain matrix for force feedback.

In this scheme:

- The force sensor measures the interaction force at the end-effector;

- The force error is computed by comparing with the desired force;
- The controller generates joint torques to reduce this error.

Characteristics and Challenges

Unlike impedance control, where force is indirectly shaped through motion, direct force control explicitly closes a feedback loop on the force.

However, this approach introduces several challenges:

- Force measurements are often noisy and may not allow reliable derivative action;
- The system stability strongly depends on the environment stiffness;
- The controller typically requires damping through velocity feedback;
- Force control is meaningful only along directions where contact forces exist.

For these reasons, force control is often implemented as an outer loop around a motion control system, such as inverse dynamics control, rather than as a standalone control input.

8.6.2 Force Control with Inner Position Loop

As discussed previously, direct force control can be implemented by closing a force feedback loop. However, directly injecting force control at the torque level may lead to stability issues, especially in the presence of stiff environments.

An alternative approach is to exploit the existing position control loop and regulate force indirectly by modifying the position reference. This leads to the concept of *force control with an inner position loop*.

We consider an inverse dynamics control law with force measurement:

$$u = D(q)a_q + C(q, \dot{q})\dot{q} + G(q) - J^T(q)F_e. \quad (8.43)$$

The control input a_q is designed in operational space as:

$$a_q = J^{-1}(q)M_d^{-1} \left(-K_D\dot{x} + K_P(x_F - x) - M_d\dot{J}(q, \dot{q})\dot{q} \right), \quad (8.44)$$

where:

- x is the end-effector position,

- x_F is a virtual reference position related to the force error,
- M_d, K_D, K_P are desired inertia, damping, and stiffness matrices.

Substituting the control law into the system dynamics yields the following *closed-loop dynamics*:

$$M_d \ddot{x} + K_D \dot{x} + K_P x = K_P x_F. \quad (8.45)$$

This equation shows that the system behaves as a position-controlled system driving x towards the reference x_F .

To achieve force control, the reference x_F is chosen as a function of the force error:

$$x_F = C_F (F_d - F_e), \quad (8.46)$$

where C_F is a diagonal compliance matrix.

In this way:

- If the measured force is smaller than desired, x_F increases;
- If the measured force is larger than desired, x_F decreases.

Thus, the controller modifies the position reference so as to regulate the contact force.

This control scheme can be interpreted as follows:

- The inner loop performs position control;
- The outer loop adjusts the position reference based on force error;
- Force regulation is achieved indirectly through motion.

Therefore, force control is realized on top of a preexisting position control structure.

Compared to direct torque-based force control:

- The scheme is more stable, as it relies on a well-behaved position loop;
- It is less sensitive to environment stiffness;
- The force is not directly imposed, but results from the interaction.

8.6.3 Parallel Force/Position Control

The force control schemes presented so far regulate the interaction force by modifying the position reference. However, such approaches may not allow explicit control of the end-effector position along directions where motion is admissible.

In many tasks, it is desirable to simultaneously control:

- the contact force along constrained directions;
- the position along unconstrained directions.

This leads to the concept of *parallel force/position control*, where both objectives are achieved within a unified control framework.

To incorporate position control, we introduce a desired end-effector position x_d . The control input is designed as:

$$a_q = J^{-1}(q)M_d^{-1} \left(-K_D\dot{x} + K_P(\tilde{x} + x_F) - M_d\dot{J}(q, \dot{q})\dot{q} \right), \quad (8.47)$$

where:

$$\tilde{x} = x_d - x, \quad (8.48)$$

and x_F is the virtual displacement generated by the force error.

The force-related reference is defined in eq. (8.46):

Substituting the control law into the system dynamics yields:

$$M_d\ddot{x} + K_D\dot{x} + K_Px = K_P(x_d + x_F). \quad (8.49)$$

This shows that the system performs position control towards the modified reference:

$$x_{\text{ref}} = x_d + x_F. \quad (8.50)$$

Equilibrium Analysis

At steady state, the end-effector position satisfies:

$$x = x_d + C_F(F_d - F_e). \quad (8.51)$$

This equation reveals the coupling between position and force control.

The control law can be interpreted as the parallel combination of:

- a position control action driving $x \rightarrow x_d$;

- a force control action modifying the reference through x_F .

Therefore:

- Along unconstrained directions, the position x_d is achieved;
- Along constrained directions, the force F_d is regulated.

Parallel force/position control combines the advantages of both position and force control:

- It enables simultaneous regulation of motion and interaction forces;
- It adapts naturally to task constraints;
- The final behaviour depends on the compliance matrix C_F and the environment properties.

Learning Outcomes

At the end of this chapter, students should be able to:

- explain the importance of interaction control in robotic manipulation and the role of contact forces in task execution;
- describe how interaction with the environment leads to constrained motion and distinguish between indirect and direct force control strategies;
- explain how force measurements are obtained or estimated and how they are incorporated into control design;
- formulate inverse dynamics control in operational space and interpret its role as a foundation for task-space and interaction control;
- explain the concept of mechanical impedance and interpret impedance control as shaping the dynamic relationship between motion and force;
- derive impedance control laws in operational space, both with and without force measurements, and analyze their resulting behavior;
- explain the principles of direct force control and formulate force feedback control laws based on force measurements;
- compare different force control architectures, including direct force control and force control with an inner position loop, in terms of stability and implementation;
- explain how force control can be achieved indirectly through motion by modifying position references;
- describe and analyze parallel force/position control schemes for simultaneous regulation of motion and force;
- understand how control objectives are distributed across different directions in task space depending on task constraints;
- select appropriate control strategies for different interaction scenarios based on task requirements, environment properties, and sensing capabilities.